[NAME OF THE INSTITUTION]

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Department of [Department Name]

PROJECT REPORT

ON

FACE RECOGNITION BASED ATTENDANCE SYSTEM

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

in

[Branch / Programme Name]

Submitted by

[Student Name]

USN / Roll Number: [USN/Roll Number]

Under the guidance of

[Guide Name]

Department of [Department Name]

[University Name]

Academic Year: [Academic Year]

[NAME OF THE INSTITUTION]

[Address with Pin Code]

Department of [Department Name]

CERTIFICATE

Certified that the project work entitled “Face Recognition Based Attendance System” carried out by Mr./Ms. [Student Name], USN [USN/Roll Number], a bonafide student of [Institution Name], in partial fulfillment for the award of Bachelor of Engineering / Bachelor of Technology in [Branch Name] of [University Name] during the year [Academic Year]. It is certified that all corrections and suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Name & Signature of Guide	Signature of HOD	Signature of Principal
[Guide Name]	[HOD Name]	[Principal Name]

External Viva

Name of the Examiners	Signature with Date
1. [External Examiner Name] 2. [External Examiner Name]	

DECLARATION

I, [Student Name], bearing USN/Roll Number [USN/Roll Number], hereby declare that the project report entitled “Face Recognition Based Attendance System” has been carried out by me under the guidance of [Guide Name], Department of [Department Name], [Institution Name]. The work presented in this report is original to the best of my knowledge and has not been submitted previously for the award of any degree or diploma.

Place: [Place] Date: [Date]	Signature of Student [Student Name]

ACKNOWLEDGEMENT

I express my sincere gratitude to [Guide Name], Department of [Department Name], for valuable guidance, continuous support, and constructive suggestions throughout the development of this project. I am thankful to the Head of the Department, Principal, and faculty members of [Institution Name] for providing the required facilities and encouragement. I also thank my family and friends for their support during the successful completion of the project.

ABSTRACT

The Face Recognition Based Attendance System automates student attendance using Python, Flask, OpenCV LBPH recognition, SQLite, and a browser-based interface. The system enables admin login, student registration, face sample capture, model training, recognition-based attendance marking, duplicate prevention, search, dashboard viewing, and CSV export. It reduces manual effort, improves record accuracy, and provides a foundation for future hardware deployment using a Raspberry Pi camera.

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INTRODUCTION

1.1 Background

Attendance management is a routine academic activity, but manual attendance consumes classroom time and is prone to proxy entries, missing records, and transcription errors. Digital attendance systems improve reliability by recording attendance in a structured database and making the records available for search, reporting, and export.

Face recognition is a contactless biometric method that identifies a person using facial features. In this project, OpenCV's Local Binary Patterns Histograms (LBPH) recognizer is used because it is lightweight, suitable for local execution, and appropriate for an academic prototype.

1.2 Problem Statement

The problem addressed by this project is the need for a simple, local, and beginner-friendly attendance system that can register students, capture face samples, train a recognition model, mark attendance automatically, prevent duplicate entries for the same day, and allow administrators to view and export records.

1.3 Motivation

To reduce time spent on manual roll-call attendance.

To avoid proxy attendance and improve record authenticity.

To provide searchable attendance data for academic monitoring.

To build a low-cost system using open-source technologies and no paid APIs.

1.4 Objectives

Develop an admin-protected web application using Flask.

Store student and attendance details in an SQLite database.

Capture face samples using a browser webcam interface.

Train an LBPH face recognizer using OpenCV.

Recognize students and mark attendance with date and time.

Prevent duplicate attendance for the same student on the same day.

Provide dashboard, search, filtering, and CSV export features.

1.5 Scope of the Project

The present project focuses on a software-based attendance system that runs locally on a computer. It supports one administrator account, student registration, browser camera capture, OpenCV model training, attendance marking, and report export. The future scope includes extending the system into a hardware-based installation using a Raspberry Pi camera module.

1.6 Importance of Automation and Face Recognition

Automation improves speed and consistency, while face recognition provides a contactless method of identity verification. Combining both approaches helps create an attendance system that is easier to operate and more reliable than manual registers.

LITERATURE SURVEY / EXISTING SYSTEM

2.1 Manual Attendance System

Traditional attendance is recorded using paper registers or spreadsheets. Although simple, this method is time-consuming and vulnerable to proxy entries, misplaced records, and calculation errors.

2.2 RFID Attendance System

RFID systems identify students using cards or tags. They are faster than manual attendance, but the card can be shared, lost, or damaged. Additional card readers and tags also increase hardware dependency.

2.3 Biometric Fingerprint Attendance

Fingerprint systems use unique fingerprint patterns for identity verification. They offer better authenticity than cards, but they require physical contact and dedicated biometric devices.

2.4 Face Recognition Based Systems

Face recognition systems identify users from camera images. They are contactless and can be integrated with web applications. Recognition accuracy depends on lighting, camera quality, face angle, dataset quality, and the recognition algorithm used.

Table 2.1 Comparison of Attendance Systems

System	Advantages	Limitations
Manual Register	Simple and low cost	Time-consuming, proxy attendance possible
RFID	Fast and easy to scan	Cards can be shared or lost
Fingerprint	Strong biometric identity	Requires contact and special hardware
Face Recognition	Contactless and camera-based	Affected by lighting and pose

2.5 Limitations of Existing Systems

Existing methods either depend heavily on manual effort or require external hardware. The proposed system attempts to balance simplicity, affordability, and automation by using a web interface and local OpenCV recognition.

SYSTEM REQUIREMENTS AND ANALYSIS

3.1 Functional Requirements

Admin must be able to log in securely.

Admin must be able to add student details such as name, USN/roll number, department, and semester.

The system must capture face samples using the browser camera.

The system must train a face recognition model using stored samples.

The system must mark attendance after recognizing a trained student.

The system must prevent duplicate attendance on the same date.

Admin must be able to view, search, filter, and export attendance records.

3.2 Non-Functional Requirements

The interface should be clean and beginner-friendly.

The system should handle common errors using readable messages.

The application should run locally without paid APIs.

Attendance operations should be stored persistently in a database.

The design should be modular to support future hardware extension.

Table 3.1 Software Requirements

Software	Purpose
Python	Main programming language
Flask	Backend web framework
OpenCV	Face detection, dataset processing, and LBPH recognition
SQLite	Local relational database
HTML, CSS, JavaScript	Frontend user interface and browser camera capture
Browser	Access to local web application and camera permission

Table 3.2 Hardware Requirements

Hardware	Minimum Requirement
Processor	Dual-core processor or higher
Memory	4 GB RAM or higher
Camera	Laptop webcam or USB camera
Storage	At least 500 MB free space for application, database, and dataset
Future Hardware	Raspberry Pi board and Raspberry Pi Camera Module

3.3 Feasibility Study

The system is technically feasible because all core components are open-source and can be run on a standard computer. It is economically feasible because it does not use paid APIs. Operationally, it is suitable for classroom demonstrations because the admin can manage students and attendance through a simple web interface.

3.4 Use Case Explanation

The primary actor is the administrator. The administrator logs in, registers students, captures face samples, trains the model, marks attendance, views records, searches data, and exports attendance. Students interact indirectly by standing in front of the camera during registration and recognition.

3.5 Database Requirements

The database must store admin credentials, student details, and attendance records. The attendance table must enforce uniqueness for each student and date combination so that duplicate attendance is avoided.

SYSTEM DESIGN

4.1 System Architecture

The architecture follows a simple client-server pattern. The browser provides the user interface and camera capture. Flask handles routing, validation, and backend processing. OpenCV performs face detection and recognition. SQLite stores admin, student, and attendance records.

Fig. 4.1 System Architecture of Face Recognition Based Attendance System

4.2 Data Flow Design

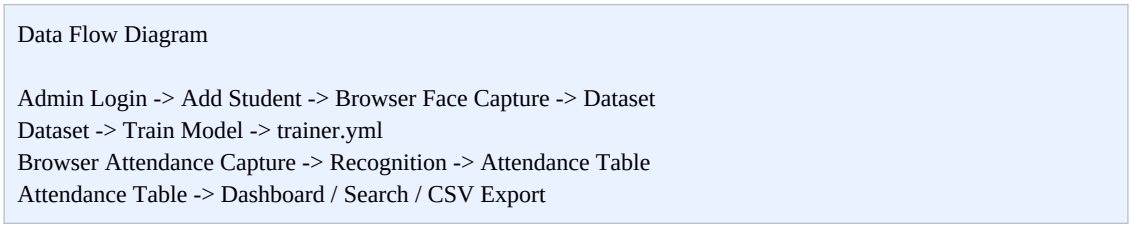

Fig. 4.2 Data Flow Diagram

4.3 Module Design

Authentication module validates admin login.

Student module stores student profile and roll number.

Capture module collects browser webcam frames.

Training module creates the LBPH model from dataset images.

Recognition module identifies faces and marks attendance.

Dashboard module displays students, daily attendance, and export features.

4.4 Database Design

```
Database Schema

admins(id, username, password_hash, created_at)
students(id, name, roll_number, department, semester, created_at)
attendance(id, student_id, attendance_date, attendance_time, status, created_at)
attendance.student_id -> students.id
UNIQUE(student_id, attendance_date)
```

Fig. 4.3 Database Schema

Table 4.1 Database Table: admins

Field	Description
id	Primary key for administrator
username	Unique admin username
password_hash	Hashed password for secure login
created_at	Record creation timestamp

Table 4.2 Database Table: students

Field	Description
id	Primary key for student
name	Student full name
roll_number	Unique USN or roll number
department	Department name
semester	Current semester
created_at	Registration timestamp

Table 4.3 Database Table: attendance

Field	Description
id	Primary key for attendance record
student_id	Foreign key linked to students table
attendance_date	Date on which attendance is marked

Field	Description
attendance_time	Time at which attendance is marked
status	Attendance status, usually Present
created_at	Record creation timestamp

4.5 Admin Workflow

Admin Workflow

Login
-> Add Student
-> Capture Face Samples
-> Train Model
-> Mark Attendance
-> View / Filter / Export Attendance

Fig. 4.4 Admin Workflow

IMPLEMENTATION

5.1 Technology Stack

The application is implemented using Python and Flask for the backend, OpenCV for face processing, SQLite for database storage, and HTML, CSS, and JavaScript for the frontend. The system uses OpenCV's LBPH face recognizer because it is lightweight and suitable for local face recognition.

5.2 Important Project Files

File / Folder	Purpose
app.py	Main Flask application and route handling
database.py	Database connection, table creation, and query helpers
train_model.py	Training LBPH recognizer from dataset images
recognize.py	Attendance marking helper functions and recognition utilities
templates/	HTML pages for login, dashboard, students, and attendance
static/	CSS and JavaScript files
dataset/	Captured face image samples
trainer/	Trained LBPH model file
attendance.db	SQLite database file
requirements.txt	Required Python dependencies

5.3 Admin Login Module

The admin login module verifies the username and password. The password is stored as a hash rather than plain text. After successful login, a session is created and protected routes become accessible.

5.4 Add Student and Browser Webcam Capture Module

The add student module accepts name, USN/roll number, department, and semester. The browser camera captures face samples and sends them to the Flask backend as

image data. The backend detects faces using a Haar cascade and stores cropped grayscale face samples in the dataset folder.

5.5 Dataset Creation and Model Training

Each captured face image is saved using the student ID in the filename. During training, the system reads dataset images, extracts student IDs, trains the LBPH recognizer, and saves the trained model as trainer.yml inside the trainer folder.

5.6 Face Recognition and Attendance Marking

For attendance, the browser captures sample frames and sends them to the backend. The trained LBPH model predicts the student identity from detected face regions. If the confidence score is within the selected threshold, the attendance helper records the student as present.

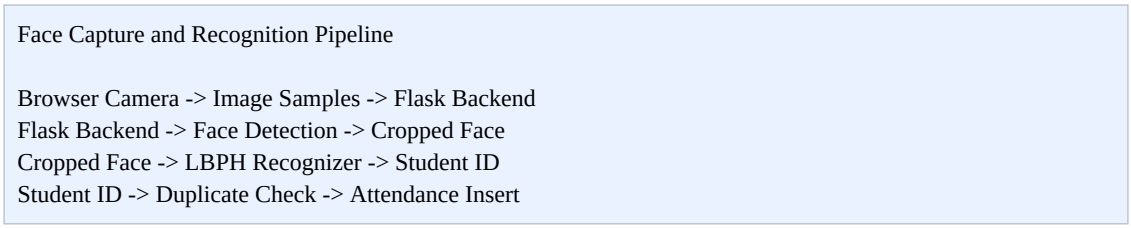

Fig. 5.1 Face Capture and Recognition Pipeline

5.7 Duplicate Attendance Prevention

The attendance table has a unique constraint on student ID and attendance date. Before inserting a record, the system checks whether the same student has already been marked on the same date. This prevents repeated entries.

5.8 Dashboard, Search, and CSV Export

The dashboard displays total students, present count for the current day, total records, and recent attendance. The attendance page supports filtering by date and student. Records can be exported as a CSV file for further processing.

5.9 Error Handling and Frontend UI

The application displays clear messages for invalid login, duplicate roll numbers, missing face samples, training errors, and attendance recognition failures. The frontend uses a clean responsive layout to make the system easy for beginners to operate.

CHAPTER 6

TESTING

6.1 Testing Approach

Testing was performed to ensure that each module works correctly and that the complete workflow from student registration to attendance export functions as expected. The system was tested using valid and invalid inputs, repeated attendance attempts, and camera capture conditions.

6.2 Types of Testing

Unit testing was used for database creation and helper functions.

Integration testing was used for login, student registration, training, recognition, and attendance storage.

Functional testing verified user-facing features such as search, filter, and CSV export.

Camera capture testing checked browser permission and sample submission.

Duplicate attendance testing verified one attendance record per student per day.

Table 6.1 Test Cases

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC-01	Admin login	Valid username and password	Dashboard opens	Dashboard opened	Pass
TC-02	Invalid login	Wrong password	Error message	Error shown	Pass
TC-03	Add student	Valid student details and face samples	Student saved	Student saved	Pass
TC-04	Duplicate roll number	Existing roll number	Friendly duplicate message	Message shown	Pass
TC-05	Train model	Dataset images	trainer.yml generated	Model generated	Pass
TC-06	Mark attendance	Recognized face samples	Attendance inserted	Attendance inserted	Pass
TC-07	Duplicate attendance	Same student same date	No duplicate record	Duplicate prevented	Pass

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC-08	Search attendance	Date or student name	Filtered records	Filtered records shown	Pass
TC-09	CSV export	Export request	CSV file generated	CSV generated	Pass
TC-10	Delete stale student	Delete action	Student removed	Student removed	Pass

6.3 Database Testing

Database testing verified that the admins, students, and attendance tables are created automatically. The unique roll number constraint and daily duplicate attendance constraint were also tested.

6.4 Camera and Recognition Testing

Camera testing verified that the browser can request camera permission, display preview, capture samples, and submit images. Recognition testing verified that trained faces can be processed and matched by the LBPH recognizer under suitable lighting conditions.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Results

The system successfully provides a working admin dashboard, student registration, browser face sample capture, model training, recognition-based attendance marking, attendance record viewing, filtering, and CSV export.

Admin can log in and access protected pages.

Students can be registered with required academic details.

Face samples are captured using browser camera frames.

The trained LBPH model is saved locally.

Attendance is marked with date and time.

Duplicate attendance is avoided for the same day.

Records can be searched, filtered, and exported.

7.2 Benefits

The system reduces manual effort, improves record organization, provides faster attendance marking, and supports searchable digital records. Because it uses open-source tools, it is cost-effective for academic learning and demonstration.

7.3 Limitations

Recognition accuracy depends on lighting, camera quality, and face position.

LBPH is suitable for prototypes but may be less accurate than modern deep learning models.

SQLite is convenient locally but should be replaced with PostgreSQL or MySQL for production deployment.

The present implementation is a software project and does not yet include dedicated hardware.

Anti-spoofing or liveness detection is not implemented in the current version.

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The Face Recognition Based Attendance System demonstrates an automated, contactless, and database-driven approach to attendance management. The project integrates Flask, OpenCV, LBPH face recognition, SQLite, and a simple frontend to support student registration, model training, recognition, duplicate prevention, dashboard viewing, and CSV export. The system is suitable as an academic prototype and can be enhanced for real-world deployment.

8.2 Future Scope / Future Extension

A major future extension is to convert the software prototype into a hardware-based attendance system using a Raspberry Pi camera. In this future version, a Raspberry Pi with a camera module can be installed at a classroom or laboratory entrance. The device can automatically capture faces, perform recognition, and mark attendance without depending on a laptop webcam.

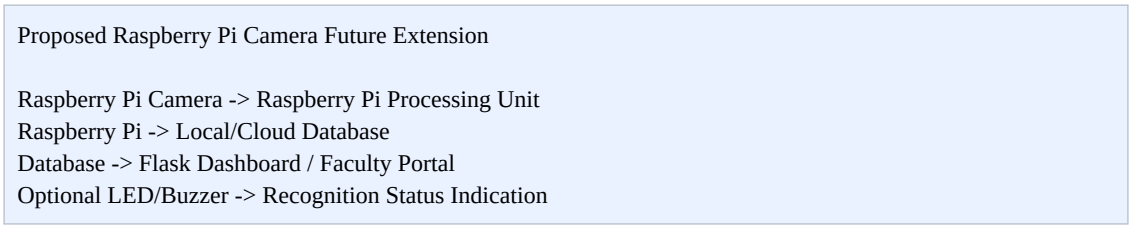

Fig. 8.1 Proposed Raspberry Pi Camera Future Extension

Raspberry Pi camera based hardware attendance system.

IoT or cloud database integration for centralized records.

Mobile application for faculty attendance monitoring.

Email or SMS alerts for attendance notifications.

Improved deep learning based face recognition.

Anti-spoofing and liveness detection.

Multi-camera support for larger classrooms.

Deployment on a cloud server with secure authentication.

Role-based access for teachers and administrators.

The Raspberry Pi hardware extension is proposed as future work only. It has not been implemented in the current software version. If implemented, the report should be expanded with complete circuit diagram, component layout, component list, and main component data sheets as required for hardware projects.

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APPENDICES

Appendix A Source Code Overview

The project source code is divided into backend files, template files, static assets, dataset storage, trainer storage, and dependency files. The modular structure makes it easier to maintain and extend.

Appendix B Installation and Setup Steps

Create a Python virtual environment.

Install required packages using pip install -r requirements.txt.

Run the Flask application using python app.py or flask --app app run.

Open the local URL in a browser.

Log in with the admin credentials.

Add students, capture face samples, train the model, and mark attendance.

Appendix C README / User Manual

The administrator should first log in, add students, capture face samples, train the model, and then use the mark attendance page. Attendance records can be viewed, filtered by date or student, and exported as a CSV file.

Appendix D Software Requirements

Package	Use
Flask	Web application framework
opencv-contrib-python	OpenCV and LBPH face recognizer
numpy	Image array processing
Pillow	Image support for training utility
Werkzeug	Password hashing and Flask dependency

Appendix E Future Hardware Extension Details for Raspberry Pi Camera

The following hardware components are proposed for a future Raspberry Pi based attendance system. These are not part of the current implementation but can be used when extending the project into a hardware prototype.

Table E.1 Proposed Raspberry Pi Hardware Components

Component	Purpose
Raspberry Pi board	Main processing unit for camera capture and recognition
Raspberry Pi Camera Module	Captures student face images
MicroSD card	Stores operating system and application files
Power adapter	Provides stable power supply
Network connection	Enables database or cloud synchronization
Optional display / LED / buzzer	Indicates recognition and attendance status
Enclosure	Protects the hardware in classroom installation

If the hardware version is implemented, actual circuit diagram, component layout, complete component list, and main component datasheets should be added to the appendix as required by the B.E./B.Tech project report guidelines for hardware projects.